A Temporal Fusion Transformer (TFT) Approach to Sector Rotation

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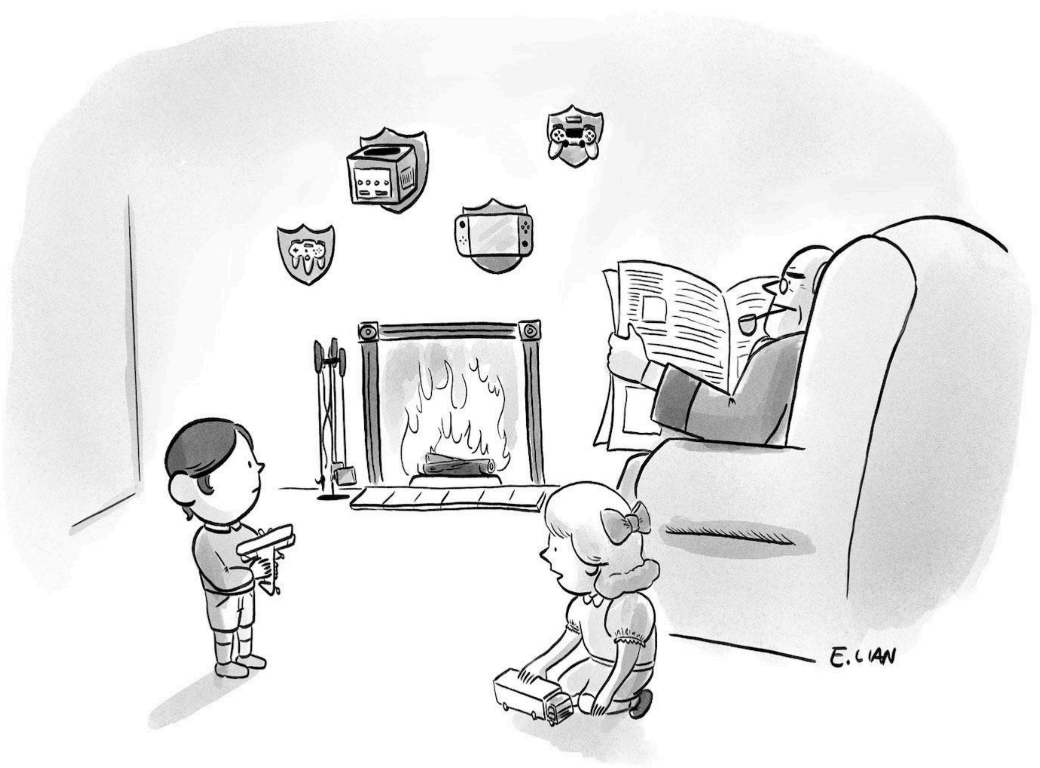

"Oh, those trophies? Father says we come from old GameStop money."

Abstract

This paper presents a robust framework for weekly sector rotation using a Temporal Fusion Transformer (TFT)[1]. Unlike standard forecasting models that predict absolute

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[1] Lim, Bryan, et al. "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting." *International Journal of Forecasting*, vol. 37, no. 4, 2021, pp. 1748-1764. https://doi.org/10.1016/j.ijforecast.2021.03.012.

price, this architecture is optimized to predict relative returns (Sector - SPY) to isolate idiosyncratic sector strength from broad market movements. We implement a walk-forward validation schema and a multi-seed ensembling approach to mitigate the high variance inherent in financial time-series data.

I. Model Architecture

The core engine is a modified TFT designed for interpretability and feature selection. The model architecture is specifically designed to handle the non-stationary nature of financial markets by incorporating several key components:

- **Variable Selection Network (VSN):** Learns which features—such as 4-week RSI, realized volatility, or relative volume—carry the most predictive weight for specific sectors at any given timestep. This allows the model to dynamically ignore noisy signals.
- **Encoder-Decoder Temporal Logic:** Employs an LSTM encoder to capture sequential momentum and a self-attention mechanism to identify critical historical weeks within a 12-week lookback period.
- **Output Head:** Instead of a single regression point, the model employs per-sector output projections to generate a full cross-sectional ranking of the 11 S&P 500 sectors.

II. Feature Engineering

The model ingests ten features per variate at each timestep, producing an input tensor of shape (lookback length × 12 variates × 10 features) for each weekly observation. The feature set was designed to provide diverse signal types—directional, volatility-based, volume-based, momentum, and cross-sectional—so that the Variable Selection Network has meaningful candidates to weight differently across sectors and market regimes.

- **Returns (returns, log_returns, open_close_return):** Weekly close-to-close simple and log returns are both included. Simple returns are the natural unit for portfolio mathematics; log returns are statistically better behaved and more nearly normally distributed, which benefits gradient-based training. Open-to-close return is also computed to capture intra-week directional pressure independent of weekend gap effects.
- **Volatility (weekly_range, realized_vol_4):** Weekly high-low range and 4-week realized volatility provide complementary risk signals. Range captures immediate uncertainty; realized volatility identifies the prevailing volatility regime. Sector rotation patterns differ markedly between high-volatility and low-volatility environments, and explicit volatility features help the model distinguish them.

- **Volume (log_volume, relative_volume):** Log-transformed volume handles the wide dynamic range of trading activity. Relative volume (current volume divided by a recent baseline) flags unusual activity. Volume confirmation is a long-established quantitative finance principle: price moves accompanied by elevated volume tend to be more durable than moves on thin volume, and relative-volume spikes often precede sustained trends.
- **Momentum (rsi_4, price_vs_sma4):** 4-week Relative Strength Index and price-versus-4-week-SMA (Simple Moving Average) both quantify recent directional strength. RSI provides a bounded oscillator sensitive to overbought and oversold conditions; the SMA ratio provides a continuous trend indicator. The 4-week window is short enough to react to regime shifts but long enough to filter out single-week noise.
- **Cross-Sectional Rank (return_rank):** The rank of each sector's recent return relative to the other ten sectors is included as a direct input. This feature is the most closely matched to the prediction target: because the model is predicting relative outperformance, providing the current relative-performance ordering accelerates learning and reduces the burden on the model to infer cross-sectional structure from raw returns alone.

All features are computed independently for SPY and for each of the eleven sector ETFs, then z-score normalized using statistics computed strictly from the training portion of each fold to prevent look-ahead leakage. Missing values are tolerated up to a maximum ratio of 5% per ticker; tickers exceeding this threshold are excluded from the relevant fold.

III. Objective Function: The Ranking Advantage

The choice of objective function is a key methodological decision in this study, and the project evolved through two formulations. The model was initially trained with Mean Squared Error (MSE) on predicted vs. actual relative returns. MSE is the standard choice for regression problems and is the most intuitive loss for a model that emits numerical predictions. However, examination of the training dynamics revealed a pathology specific to the small-magnitude nature of weekly relative returns: MSE collapsed trivially toward zero (training loss fell by a factor of 600 within 100 epochs) while validation rank correlation crawled to only 0.115. The optimizer was driven primarily by predicting numbers near zero, providing weak gradient signal for the cross-sectional ranking task that the strategy actually depends on.

The model was subsequently retrained with a RankNet[2]-style pairwise ranking loss. For each pair of sectors (i, j) in each training week, the loss penalizes the model when the predicted ordering pred[i] vs. pred[j] disagrees with the actual ordering target[i] vs. target[j]:

L_pair = softplus(-(pred[i] - pred[j]) · sign(target[i] - target[j]))

Pairs with effectively-tied targets are excluded from the loss. The implementation is fully vectorized, producing all valid pairs per batch in a single tensor operation. This loss directly optimizes for the ranking quality that the long/short strategy depends on, rather than for absolute return accuracy.

In a sector rotation strategy, the absolute predicted return is less critical than the accuracy of the ranking. By maximizing the correlation between predicted and actual sector ranks, the model is incentivized to correctly identify the leaders and laggards, which is the mechanical requirement for a long/short strategy. The empirical impact of this change is documented in Section V.

Output Calibration for Display

A direct and important consequence of the ranking objective is that the model's outputs are calibrated in their ordering but not in their magnitude. The pairwise loss rewards the model for placing sectors in the correct relative order; it imposes no penalty for the absolute scale of the predicted values. As a result, the raw model outputs can be roughly an order of magnitude larger than realistic weekly sector relative returns. In one representative week the raw predictions ranged from approximately +0.22 to −0.23, which, if interpreted literally as returns, would imply implausible weekly moves of +22 to −23 percent. This poses no problem for the trading strategy itself, which consumes only the ordering of the predictions to select the top-three and bottom-three sectors, and is therefore invariant to any rescaling of the outputs. It does, however, present a problem for the public-facing presentation of the picks on the MarketMaker web interface, where the raw values were displayed as predicted percentage returns and consequently appeared implausible.

To present sensible figures without altering the strategy, a post-hoc linear calibration is applied to the predictions before they are published. The full cross-section of eleven sector predictions for the week is rescaled by a single multiplicative factor so that its cross-sectional standard deviation matches a target representative of real weekly sector relative returns, set to 1.5 percent. Because the same factor is applied uniformly to every sector, the transformation is strictly monotonic: it preserves the ordering of the

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[2] Burges, C., Shaked, T., Renshaw, E., Lazier, A., Deeds, M., Hamilton, N., & Hullender, G. (2005). Learning to rank using gradient descent. *Proceedings of the 22nd International Conference on Machine Learning - ICML '05*, 89-96. https://doi.org/10.1145/1102351.1102363

predictions exactly, and therefore leaves the long and short selections completely unchanged. Only the displayed magnitudes are affected. In the representative week noted above, this calibration mapped the raw predictions to a realistic range of approximately +2.5 to −2.6 percent while retaining the identical ranking, and hence the identical trades.

It is worth emphasizing what this calibration is and is not. It is a presentation-layer transformation that maps the model's ordinal scores onto an interpretable return scale for human readers; it is not a correction to the model, nor does it influence any backtested or live performance figure. All Sharpe ratios, win rates, and walk-forward results reported in this paper are computed from realized strategy returns and are entirely independent of how the prediction scores are displayed. The calibration target is a single configurable parameter and can be adjusted to taste without any effect on the underlying strategy.

IV. Strategy Execution & Walk-Forward Simulation

To ensure the findings are not artifacts of overfitting, we utilize a Walk-Forward validation cycle. This simulates a real-world trading environment by ensuring the model never sees data from the future during training.

- **Training Routine:** The model is trained on an 8-year rolling window (~400 weeks) and validated on a 1-year period, followed by testing on the subsequent 6 months (~26 weeks).
- **Market-Neutral Posture:** The strategy goes long the top 3 predicted outperformers and short the bottom 3 underperformers.
- **Ensembling:** Predictions are averaged across multiple random seeds (e.g., 5 seeds per fold) to stabilize the attention weights.

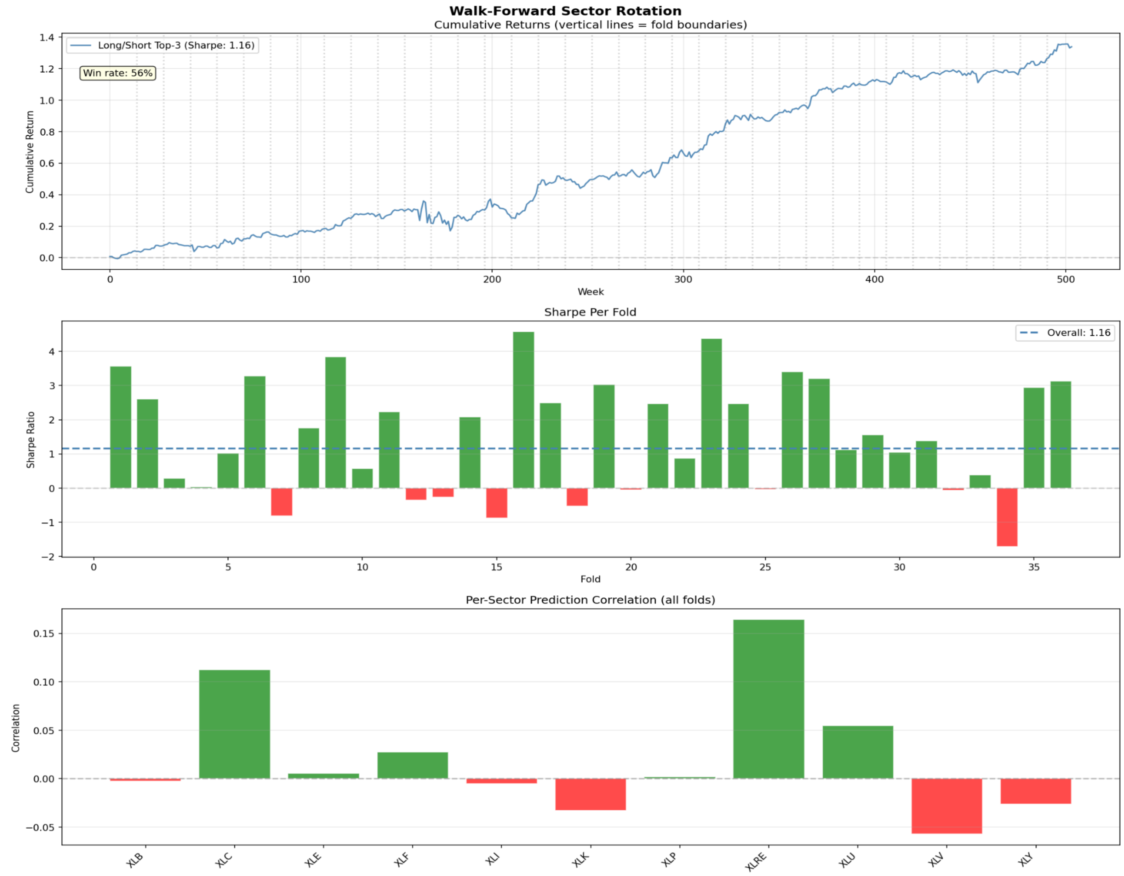

Figure: Walk-forward evaluation of the long/short sector rotation strategy across 37 folds spanning approximately 500 weeks (roughly ten years out-of-sample). Top: cumulative return of the top-3 long / bottom-3 short strategy, reaching approximately 1.35 at an aggregate Sharpe ratio of 1.16 and a 56% weekly win rate. Middle: per-fold Sharpe ratios (green positive, red negative), with 28 of 37 folds positive. Bottom: per-sector prediction correlation aggregated across all folds, strongest for XLRE and XLC.

V. The Sharpe Ratio: Quantifying Risk-Adjusted Performance

For a market-neutral sector rotation strategy, the Sharpe ratio[3] serves as the primary performance metric. It measures the magnitude of excess returns relative to their volatility, which is the relevant question for a strategy designed to generate steady, uncorrelated alpha rather than to amplify market beta.

- **Mathematical Form:** Sharpe = (mean return − risk−free rate) / standard deviation of returns. For weekly returns, the value is annualized by multiplication by $\sqrt{52}$, giving comparability with strategies operating at other frequencies.

[3] “Sharpe Ratio.” *Wikipedia, The Free Encyclopedia*, Wikimedia Foundation, https://en.wikipedia.org/wiki/Sharpe_ratio. Accessed 15 May 2026.

- **Why It Suits This Project:** Because the long/short construction removes broad-market exposure, the strategy is expected to produce a modest but persistent return stream. The Sharpe ratio rewards exactly this profile—consistency over magnitude—and is the metric institutional allocators expect to see for any market-neutral strategy.
- **Interpretation Bands:** Sharpe below 0.5 is typically considered weak or indistinguishable from noise; 0.5 to 1.0 reflects the historical range of major equity risk factors after costs; 1.0 to 2.0 covers most successful systematic equity strategies; values sustained above 3.0 on meaningful capital are vanishingly rare and warrant skepticism.
- **Single-Seed vs. Ensemble Reporting:** A single-seed walk-forward Sharpe is biased upward as an estimator of expected out-of-sample performance, since selecting the best result among multiple random initializations does not generalize to future runs. The ensemble mean across multiple seeds provides the more honest expected value, while the standard deviation across seeds quantifies the uncertainty around that estimate.
- **Limitations:** The Sharpe ratio assumes returns are approximately normally distributed and penalizes upside volatility identically to downside volatility. The Sortino ratio addresses the latter concern by penalizing only downside variance. Despite these limitations, the Sharpe ratio remains the de facto standard in quantitative finance for risk-adjusted performance evaluation, and is reported alongside cumulative return and maximum drawdown in any rigorous strategy evaluation.

Following correction of the loss function (transition from MSE to pairwise ranking loss, described in Section III) and a corresponding fix to model checkpoint selection (saving the model that maximized validation rank correlation rather than validation loss), a fresh single-training-run evaluation was performed using a 70/15/15 train/validation/test split. This evaluation served as a verifiable baseline benchmark for the corrected methodology. Three configurations were evaluated as a controlled ablation:

- **MSE baseline:** test rank correlation 0.095, test top-3 accuracy 34.8%. Best epoch 91 of 100 (training had not yet converged at termination).
- **Pairwise ranking loss:** test rank correlation 0.138, test top-3 accuracy 36.0%. A 45% relative improvement in rank correlation over the MSE baseline.
- **Pairwise loss with rank-correlation-based checkpoint selection:** test rank correlation 0.170, test top-3 accuracy 34.5%. An additional 23% relative improvement over the previous configuration, and 79% total improvement over the MSE baseline.

The progression demonstrates that each methodological refinement contributed measurable improvement, with the loss-function change being the largest single contributor. The final test rank correlation of 0.170 corresponds, by proportional scaling from the MSE-baseline strategy Sharpe of approximately 0.50 observed on the same evaluation split, to an expected Sharpe ratio in the range of 0.75 to 1.0—substantively improved over the baseline but more conservative than the preliminary walk-forward figure. Per-sector predictive correlation in this evaluation showed positive signal on 7 of 11 sectors, with the strongest contributors being XLC (correlation 0.31) and XLRE (0.29). The broader distribution of predictive signal across sectors, relative to the XLRE-concentrated pattern observed in the preliminary walk-forward, suggests the corrected training pipeline is extracting genuine cross-sectional ranking signal rather than relying disproportionately on a single sector.

Definitive Evaluation

The definitive evaluation combines all methodological improvements—hybrid pairwise-plus-MSE loss for both ranking quality and scale calibration, macroeconomic features (Treasury, credit, dollar, and volatility variates added alongside the eleven sector ETFs), rank-correlation-based checkpoint selection, and increased dropout regularization—in a 10-seed ensemble walk-forward spanning 37 folds and approximately 500 weeks, roughly ten years of strictly out-of-sample testing. This evaluation produced an aggregate Sharpe ratio of 1.16 with a 56% weekly win rate; approximately 76 percent of folds (28 of 37) were positive. Per-fold Sharpe ratios ranged from roughly −1.7 to +4.6, a far tighter and more stable distribution than the earlier small-sample evaluations. The cumulative return curve climbed steadily from zero to approximately 1.35 over the full period, traversing multiple distinct market regimes—including the 2018 selloff, the 2020 pandemic crash, the 2022 bear market, and the 2023–2024 recovery—without extended drawdowns.

This figure is more conservative than several smaller-sample evaluations conducted during development, but it is substantially more credible. The ten-year, 37-fold sample provides far narrower confidence bounds, and the steady equity curve across heterogeneous regimes indicates a persistent structural edge rather than a small number of favorable folds; a per-fold view still shows meaningful regime sensitivity, with individual folds ranging from mildly negative to strongly positive, which is expected for any directional sector strategy. The per-sector predictive correlations, while lower in magnitude than single-period estimates (XLRE 0.16, XLC 0.11), reflect the honest long-run signal strength after averaging across a decade of varied conditions; a single fold may exhibit a much stronger XLRE correlation, but the decade-averaged figure is the appropriate basis for forward expectations. Accounting for the anticipated 15 to 25 percent degradation from real-world frictions—transaction costs, slippage, and option bid-ask spreads—a realistic deployed Sharpe ratio is estimated in the range of 0.9 to 1.0, which remains a solid risk-adjusted return for a market-neutral strategy.

This 1.16 decade-spanning result, rather than any of the higher small-sample figures, is the headline performance characterization of the strategy. It is the most reproducible and least sample-dependent number available, and the figure least likely to disappoint under out-of-sample deployment.

VI. Model Interpretability: Variable Selection

A central design feature of the Temporal Fusion Transformer is interpretability through its Variable Selection Network. At every timestep the network emits a softmax distribution over the input variates, and because these weights are normalized to sum to one, they can be read directly as the proportion of the model's attention allocated to each input. Averaging these weights across all samples and timesteps yields a global measure of which inputs the model relies upon. The analysis below was produced by capturing the variable selection weights from the trained model across the full evaluation history.

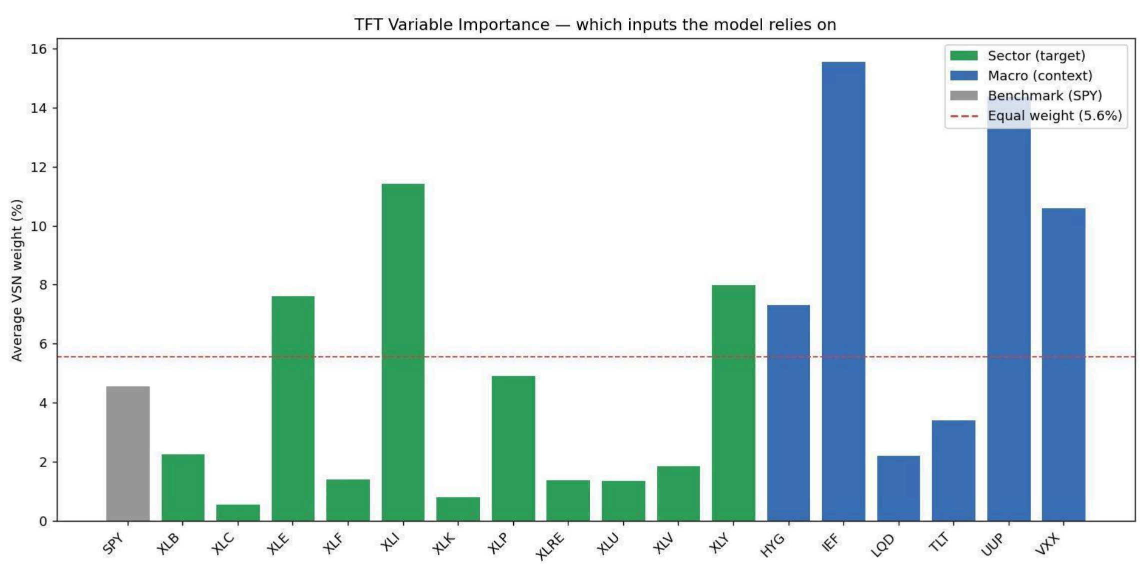

Figure: Average Variable Selection Network weight per input variate. Green bars are sector ETFs (prediction targets), blue bars are macroeconomic context variates, gray is the SPY benchmark. The dashed line marks equal weight (5.6%).

The complete ranking underlying the chart is given in the table below, ordered by weight and annotated with each variate's group and a proportional bar. The group totals at the foot of the table summarize the headline finding: macroeconomic context variates collectively command the majority of the model's attention.

VARIABLE IMPORTANCE (avg VSN weight, all samples)

Rank	Ticker	Group	Weight	
1	IEF	macro	15.58%	###############################
2	UUP	macro	14.38%	#############################
3	XLI	sector	11.44%	#######################
4	VXX	macro	10.61%	#####################
5	XLY	sector	8.02%	################
6	XLE	sector	7.63%	###############
7	HYG	macro	7.33%	###############
8	XLP	sector	4.93%	##########
9	SPY	benchmark	4.60%	#########
10	TLT	macro	3.44%	#######
11	XLB	sector	2.29%	#####
12	LQD	macro	2.24%	####
13	XLV	sector	1.87%	####
14	XLF	sector	1.44%	###
15	XLRE	sector	1.40%	###
16	XLU	sector	1.39%	###
17	XLK	sector	0.84%	##
18	XLC	sector	0.58%	#

Group totals: sectors 41.8% | macros 53.6% | benchmark 4.6%

Table: Complete variable selection weights for all eighteen input variates, ranked by average attention. Intermediate Treasuries (IEF), the US dollar (UUP), and equity volatility (VXX) occupy three of the top four positions; the sectors with the strongest walk-forward predictive correlation (XLRE, XLC) rank near the bottom in direct weight, indicating they are forecast from macroeconomic context rather than their own price histories.

The result is striking: the model allocates the majority of its attention to macroeconomic context rather than to the sector price histories it ultimately ranks. The six macro variates collectively receive 53.6 percent of total weight, the eleven sectors receive 41.8 percent, and the SPY benchmark receives 4.6 percent. The three highest-weighted individual inputs in the entire model are intermediate Treasuries (IEF, 15.58 percent), the US dollar (UUP, 14.38 percent), and equity volatility (VXX, 10.61 percent); together with high-yield credit (HYG, 7.33 percent), these four macro signals account for roughly 48 percent of the model's attention.

This reveals an economically coherent strategy that the model discovered on its own. It reads the macroeconomic regime—the level of intermediate rates, the strength of the dollar, the state of equity volatility, and the condition of credit markets—and uses a small number of bellwether cyclical sectors as confirmation, principally Industrials (XLI, 11.44 percent), Consumer Discretionary (XLY, 8.02 percent), and Energy (XLE, 7.63 percent). It then ranks all eleven sectors off that macro read. This is closely analogous to how a discretionary macro strategist would approach sector rotation, and the Variable Selection Network arrived at the structure without explicit instruction.

A notable inversion supports this interpretation. The sectors that exhibited the strongest predictive correlation in walk-forward evaluation—Real Estate (XLRE) and Communication Services (XLC)—received the lowest direct variable selection weight, at 1.40 percent and 0.58 percent respectively. The model is therefore not forecasting these sectors from their own price histories; it is forecasting them from the shared macroeconomic drivers. Rate-sensitive and defensive sectors in particular appear to be predicted indirectly through the macro variates rather than read directly, which is consistent with the economic reality that such sectors are largely driven by the rate and risk environment.

The decomposition also exposes redundancy that informs future model design. Within the rates complex, intermediate Treasuries (IEF) dominate long Treasuries (TLT, 3.44 percent) by more than fourfold, indicating that the belly of the yield curve carries substantially more sector-rotation signal than the long end. Within credit, high-yield (HYG) dominates investment-grade (LQD, 2.24 percent). A leaner macro set retaining only IEF, UUP, VXX, and HYG would likely preserve most of the model's informational advantage while reducing parameter count and concentration. Conversely, the heavy reliance on six macro series—over half of total attention—represents a concentration that warrants monitoring, as the strategy is implicitly exposed to those series behaving in the future as they did historically.

VII. Real-World Friction & Option Modeling

The strategy execution involves a simulation engine that accounts for variables often ignored in academic backtests, bridging the gap between theoretical alpha and tradable returns.

- **Derivative Implementation:** The model simulates buying $1.00 OTM calls for long picks and OTM puts for shorts, with roughly 30 days to expiration.
- **Friction and Decay:** The simulation includes per-contract commissions and accounts for time decay (Theta) and Bid/Ask spreads.

VIII. Conclusion

The Temporal Fusion Transformer provides a resilient signal for sector rotation by focusing on cross-sectional ranking and utilizing an interpretable architecture. Future research will focus on incorporating alternative data sets, adding multi-horizon momentum features, and conducting a systematic hyperparameter search. The hyperparameter optimization capability (via Optuna) is implemented in the codebase but was deliberately not applied to the results reported here; all figures reflect a manually specified configuration, and tuning is reserved as a distinct avenue for future improvement so that any resulting gains can be cleanly attributed.

Appendix: Command-Line Interface and Execution Flags

1. Core Pipeline (run.py)

Main entry point to execute the full end-to-end lifecycle.

- **--skip-download:** Bypasses yfinance data retrieval.
- **--skip-preprocess:** Skips feature engineering and loads existing tensors.
- **--epochs [INT]:** Overrides default training duration.
- **--device [STR]:** Forces execution on a specific device (cpu, cuda, mps).
- **--tune:** Enables Optuna hyperparameter search prior to final training. Not used for the results reported in this paper; reserved for future work.

2. Backtesting and Validation (walk_forward.py)

- **--train-weeks [INT]:** Weeks in the rolling training window (default 400).
- **--val-weeks [INT]:** Size of the validation window (default 52).
- **--test-weeks [INT]:** Size of the out-of-sample test window (default 26).
- **--n-seeds [INT]:** Model initializations to ensemble per fold.

3. Strategy Simulation (simulation.py)

Command: setup

- **--capital [FLOAT]:** Starting cash balance.
- **--otm [FLOAT]:** Strike distance from current price.
- **--commissions [FLOAT]:** Fee per contract per side.

Command: predict, execute, status, close, week, reset